



CP302

CAPSTONE PROJECT – 1

Topic:

Development of a Bone Implant

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Index

1. Abstract
2. Chapter – 1. Introduction
 - 1.1. Objectives
 - 1.2. Problem description
3. Chapter – 2. Exciting Studies and Literature Review
4. Chapter – 3. Working Mechanism and Design
 - 3.1. Key Components
 - 3.2. Expected Outcomes
5. Chapter – 4. Control System
 - 4.1. Components of control system
 - 4.2. PID control of the plant
6. Future Aims
7. Conclusion
8. References

1. Abstract

Limb-lengthening surgeries have significantly evolved from traditional external fixation methods to advanced internal implants, improving patient comfort and recovery. However, high costs and limited accessibility hinder their widespread use, particularly in developing nations like India. This project aims to develop a cost-effective limb-lengthening nail system utilizing a motor-driven linear actuator, a planetary gear system, and a lead screw mechanism. By leveraging additive manufacturing and localized production methods, we seek to create an affordable alternative to existing models like Fitbone and Precice Nail while maintaining precision, durability, and ease of control. Our approach focuses on optimizing mechanical design to ensure controlled bone distraction, minimizing energy consumption, and integrating non-invasive external control for patient and physician convenience. If successful, this innovation could make limb-lengthening procedures more accessible, offering a transformative solution for orthopaedic treatment in cost-sensitive healthcare environments.

Chapter-1. Introduction

Limb-lengthening is a medical procedure used to gradually increase the length of bones. The process is based on the biological principle of distraction osteogenesis, where a bone is surgically cut and then slowly pulled apart, allowing new bone tissue to grow and fill the gap.

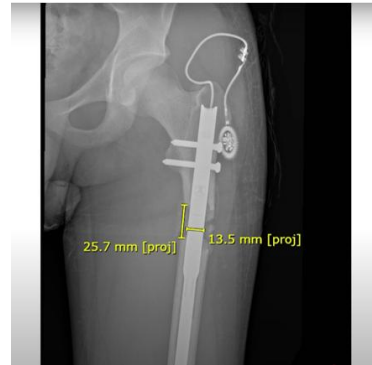
There are two main types of limb-lengthening procedures:

- 1. External fixation methods:** External fixation methods are the traditional approach, using an external frame attached to the bone with pins and wires. These methods are cost-effective and widely used, especially in developing countries. However, they have several disadvantages, including a high risk of infection, pain, scarring, and long treatment durations due to the prolonged use of external hardware.



The image shows a limb-lengthening procedure using an external fixation device.

- 2. Internal lengthening nails:** Internal lengthening nails (implants) are modern, implantable devices placed inside the bone, offering a less invasive and more comfortable alternative. Examples include Fitbone, which uses a motorized system, and Precice Nail, which relies on a magnet-based mechanism. These implants have lower infection risks, cause less pain, and provide better patient comfort compared to external fixators. However, they are significantly more expensive and require advanced surgical expertise, making them less accessible in certain regions.



Images of internal limb-lengthening system inside a bone.

1.1. Objectives

The primary objectives of this project are:

- To create a low-cost Limb-Lengthening Nail substitute by combining a lead screw, an improved planetary gear system, and a motor-driven linear actuator.
- To guarantee that, through exacting mechanical design and testing, the system retains accuracy and robustness under physiological demands.
- To provide a non-invasive external control system that allows for mechanical or wireless adjustments for the convenience of patients and doctors.
- Using knowledge from previous research on distraction osteogenesis and mechanical stress in bone healing, the mechanical design will be validated.
- To lower costs and promote local production by utilizing domestic manufacturing methods including cost-effective machining and additive manufacturing (3D printing).

1.2. Problem Description

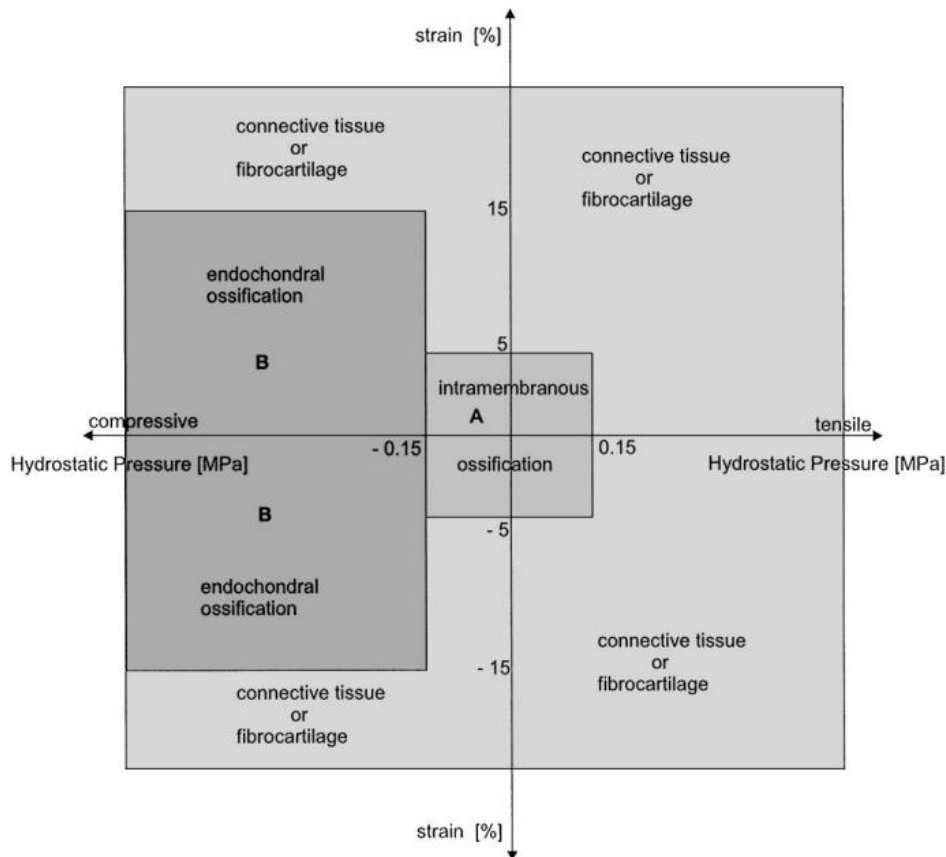
The high-end materials, complex electronic components, and precise control mechanisms used in existing Limb-Lengthening Nail systems make them costly. To guarantee dependability and safety, these systems need to be thoroughly researched, precisely engineered, and manufactured using premium techniques. The following are some major obstacles to creating a financially viable substitute:

- To avoid issues like non-union or undue stress on the bone structure, the device must permit regulated, progressive bone elongation.
- The apparatus needs to be able to withstand the mechanical strain caused by daily activities and body weight.
- Patients and doctors should be able to externally regulate the lengthening process without the need for intrusive treatments.
- The challenge is to use cost-effective materials and manufacturing techniques without compromising functionality.

Chapter-2. Existing Studies and Literature Review

Limb-lengthening implants like Fitbone and Precice Nail are advanced medical devices used to gradually increase bone length in cases of uneven limbs or for height enhancement. Fitbone, made by Orthofix, is a fully implanted device with a small motor inside. It is controlled externally using a remote, allowing precise bone stretching while reducing infection risks compared to older external methods. The Precice Nail, by NuVasive, works similarly but uses a magnet-based system instead of a motor. A magnetic controller placed outside the body activates the implant's internal gears to lengthen the bone without surgery. These modern implants offer benefits like less pain, fewer infections, and more comfort compared to traditional external fixators. However, their high cost and limited availability make them inaccessible to many people, especially in developing countries. This is why affordable alternatives, like the one in our project, are important.

➤ Stress-strain conditions for bone formation:



From the **diagram** (strain vs. hydrostatic pressure), the following observations can be made:

1. Endochondral Ossification (B Region)

- Occurs under high compressive hydrostatic pressure (>0.15 MPa).
- Happens when strain is moderate (~ 5 -15%).
- Seen in fracture healing with flexible fixation, where movement allows callus formation.

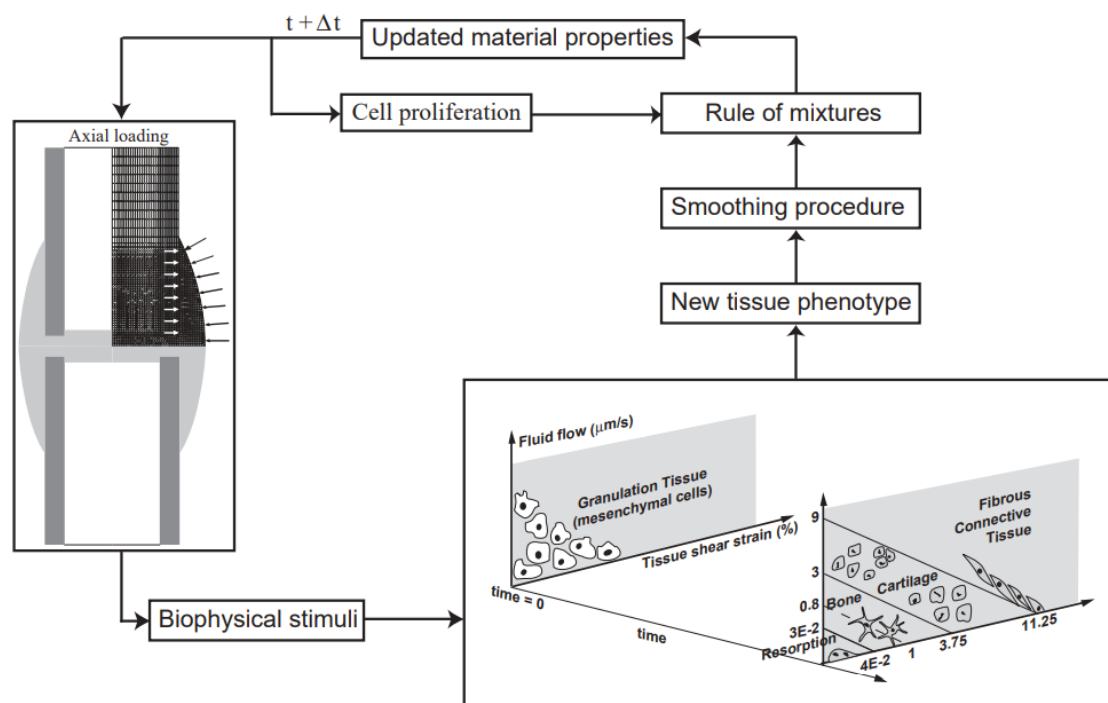
2. Intramembranous Ossification (A Region)

- Occurs at low strain (<5%) and low hydrostatic pressure.
- Seen in fracture healing with rigid fixation, where minimal movement allows direct bone formation.

3. Connective Tissue/Fibrocartilage Formation

- High strain (>15%) or excessive hydrostatic pressure prevents ossification.
- This is undesirable as it leads to non-union or delayed healing.

➤ Biophysical Stimuli and Tissue Differentiation



- The simulation accounts for **axial loading** on a healing fracture, which generates **fluid flow and shear strain** in the fracture callus.
- Depending on these mechanical conditions, different **tissue phenotypes** emerge, as indicated in the bottom-right graph.

Mechano-Regulation of Tissue Formation

- **Endochondral ossification (Cartilage Formation & Replacement by Bone)**
 - If shear strain is moderate and fluid flow is high, MSCs differentiate into granulation tissue and then cartilage.
 - This cartilage later undergoes resorption and is replaced by bone through endochondral ossification.

- This aligns with moderate strain and compressive stress, as shown in the previous stress-strain diagram.
- **Intramembranous ossification (Direct Bone Formation)**
 - If shear strain is very low ($<1\%$) and fluid flow is low, MSCs directly differentiate into bone without forming a cartilage intermediate.
 - This corresponds to rigid fixation conditions, leading to intramembranous ossification (as seen in the previous diagram's central region).

Iterative Healing Process

- The **updated material properties** step in the flowchart indicates that the fracture site **stiffens over time**, reducing strain and favouring more bone formation.
- As healing progresses, **endochondral ossification transitions into intramembranous ossification** when mechanical stability increases.

These studies will guide the design of the actuator system, ensuring that mechanical loads applied to the bone are optimized for effective regeneration.

Chapter – 3: Working Mechanism and Design

The setup is powered by a DC motor and a three-stage planetary gear system. The motor provides high-speed rotary input, which is reduced in speed and amplified in torque through the three planetary stages. This enhanced torque is then transferred to a lead screw, converting the rotary motion into precise linear movement. As the lead screw rotates, it moves the attached extruder back and forth along the actuator's axis. This setup ensures smooth, efficient, and controlled linear motion for the nail.

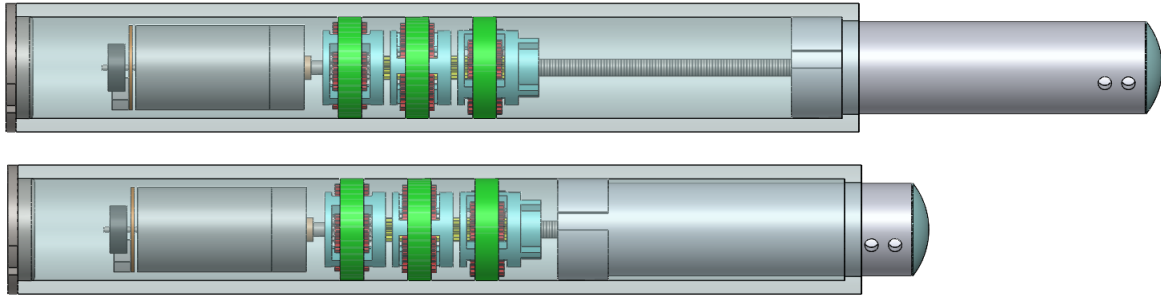


Fig. 3.1. Design of Implant Nail made in SolidWorks

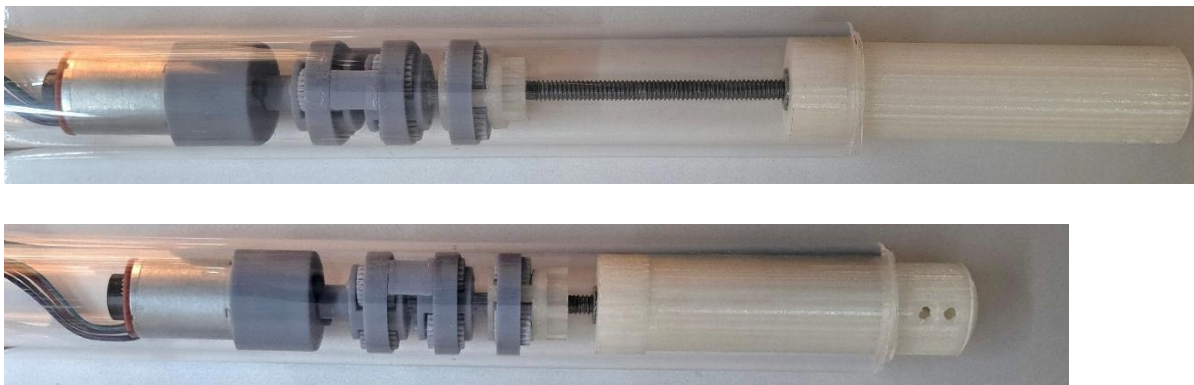


Fig. 3.2. Prototype of the model made to demonstrate the working mechanism

3.1. The working mechanism of the nail consists of the following key components:

1. **Driving motor with hall-effect encoder:** The driving motor used in this mechanism is a 6V DC motor rated at 260 RPM and equipped with a Hall-effect encoder. This motor provides the rotational input required to power the multi-stage planetary gear system. The built-in Hall-effect encoder plays a critical role in motion control by providing real-time feedback on the motor's speed and position. This allows for precise control of the motor's rotation, ensuring accurate and consistent movement of the gear system.



Fig 3.3. Motor with hall effect encoder

2. **Planetary gear system:** The planetary gear system is designed to reduce the rotational speed (RPM) while maintaining high torque efficiency. It ensures smooth and gradual transmission, making it ideal for applications requiring reliable and controlled motion.

This gear system operates using three main types of gears:

- I. **Sun Gear:** Positioned at the center of the gear assembly, the sun gear receives input directly from the motor. It has 12 teeth with a module of 0.5 mm.
- II. **Ring Gear:** This is the stationary outermost gear that encases the entire gear system. It has 54 teeth and serves as the structural boundary of the planetary setup.
- III. **Planet Gears:** Located between the sun gear and the ring gear, these gears transmit power from the sun gear to the output shaft at a reduced speed. Each planetary gear has 18 teeth.

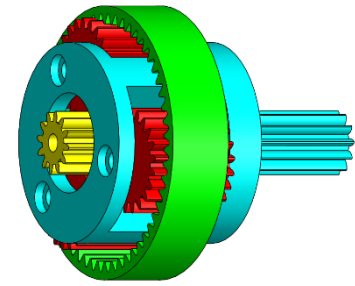


Fig. 6.4. Planetary gear system of one stage with sun gear in yellow, ring gear in green and planet gear in red in colour.

$$\text{Gear ratio of one stage of gear system} = 1 + \frac{N_r}{N_s} = 1 + \frac{54}{12} = 5.5$$

here, N_s = no. of teeth on sun gear.

N_p = no. of teeth on ring gear.

$$\text{Gear-Ratio for three stages} = (5.5)^3 : 1 = 166.375 : 1$$

3. **Lead screw mechanism:** Converts rotational motion into linear displacement with high precision, crucial for controlled bone distraction.

Pitch of lead screw = 1.25 mm per revolution.

Which linearly displace the extuder connected at the lead screw 1.25mm when lead screw rotates one full revolution. It helps to precisely control the lengthening mechanism of the nail as slow as needed.

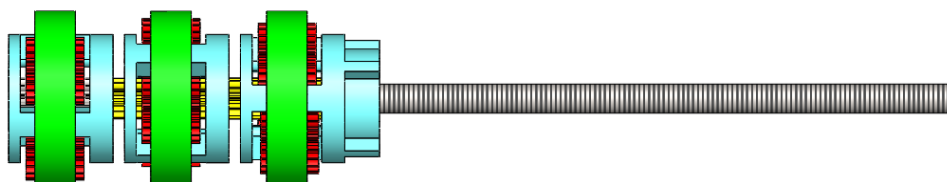


Fig. 6.5. Planetary gear system connected with lead-screw

From the pitch 1.25mm of the lead screw the linear speed of the extruder

$$= (\text{RPM of lead screw}) \times \text{pitch}$$

$$= \text{RPM} \times 1.25 \text{ mm per minute}$$

The desired linear speed of the extruder is 0.3 mm in 2 minutes.

From above we can get the required RPM of the motor to get a desired linear speed is around 50 to 80 RPM. Which can be get by PID controlling of the dc motor.

6.2 Expected Outcomes

- The actuator should provide controlled, gradual bone distraction without introducing mechanical instability.
- The planetary gear system should maintain load-bearing capacity while ensuring efficiency and minimizing energy consumption.
- The external control mechanism should enable wireless or mechanical adjustments, ensuring ease of use and non-invasive operation.
- The overall system should be significantly more affordable than existing Limb-Lengthening Nail models while maintaining performance standards.

Chapter – 4: Control System

To accurately control the position of a DC motor, a closed-loop PID control system is implemented. The system uses an Arduino Uno as the controller, an L298N motor driver to control power to the motor, and a Hall-effect encoder for feedback. The PID algorithm continuously adjusts the motor's movement until the desired position is reached with precision.

1. Components of control system:

A. Arduino Uno (Controller)

The Arduino Uno is the brain of the system. It runs a **PID control algorithm** that continuously calculates the error between the desired position (setpoint) and the current position (from encoder feedback). Based on this error, it computes a control signal to adjust motor power.

- **P (Proportional)**: Reacts to current error – large error → large correction.
- **I (Integral)**: Reacts to accumulated past errors – corrects slow drift.
- **D (Derivative)**: Reacts to rate of change – prevents overshooting.

Arduino reads the encoder pulses via interrupt pins, calculates position in real time, compares it with the target position, and sends a PWM signal to the L298N accordingly.

B. L298N Motor Driver

The L298N acts as a bidirectional driver that allows Arduino (which operates at low current) to control the higher-current DC motor.

- **IN1 & IN2**: By changing high and low we can control the direction of rotation.
- **ENA**: Takes PWM signal from Arduino to control motor speed.
- **H-Bridge Structure**: Allows motor to run forward or reverse based on logic levels of IN1 and IN2.

The Arduino sets the direction pins based on desired position (e.g., clockwise for forward, counterclockwise for reverse), and adjusts the PWM duty cycle to control speed based on PID output.

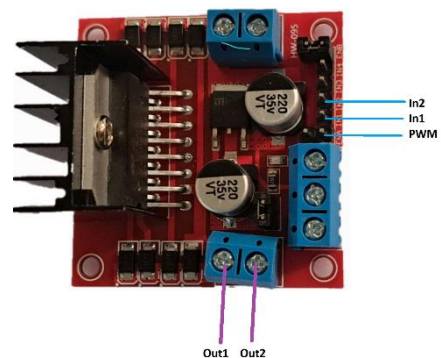


Fig. 4.1. L298N motor driver

C. Hall-effect Encoder (Feedback Sensor)

The Hall-effect encoder is mounted on the motor shaft and generates a series of digital pulses as the motor rotates.

The encoder is consisting of a hall-effect sensor with reads the changes in magnetic field by the magnet attached with shaft.

The cables connected to encoder are:

- i. Encoder (-): Ground signal into the encoder
- ii. Encoder A: Gives the number of pulses sensed by the hall-effect sensor.
- iii. Encoder B: Gives the rotating direction of the motor shaft.
- iv. Encoder (+): 5V into the power the encoder.

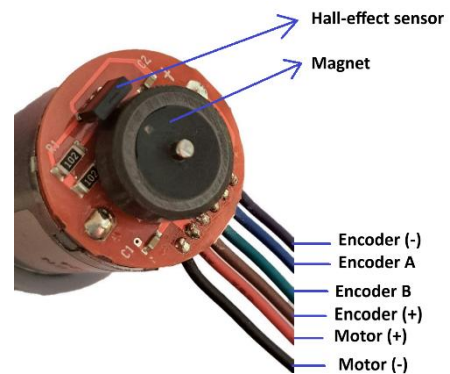


Fig. 7.3 Encoder of the Motor

Each pulse corresponds to a small angular movement. Counting pulses tells how far the motor has rotated. Pulse frequency gives speed, and pulse count gives position. Arduino reads these pulses using interrupt service routines (ISR), ensuring no pulses are missed even at high RPM. The total pulse count is then translated to position (in degrees).

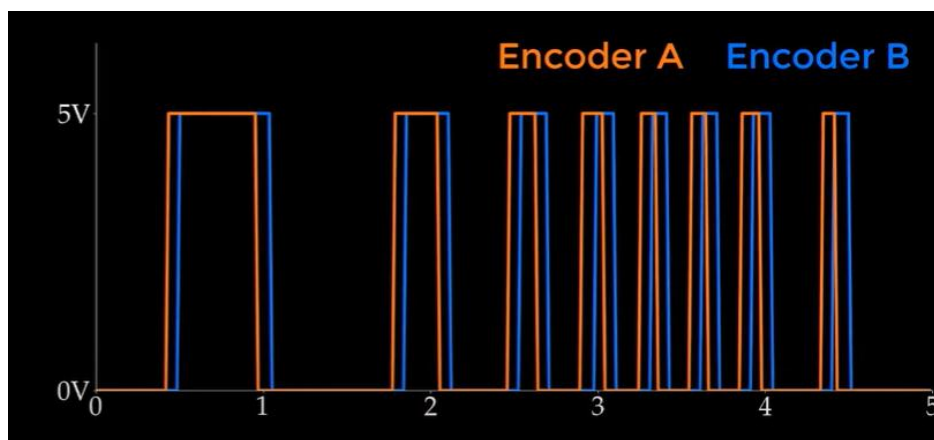


Fig. 4.3. Image of pulses generated by hall-effect sensor and encoder A gives the number of pulses generated and B gives the direction

2. PID control of the plant:

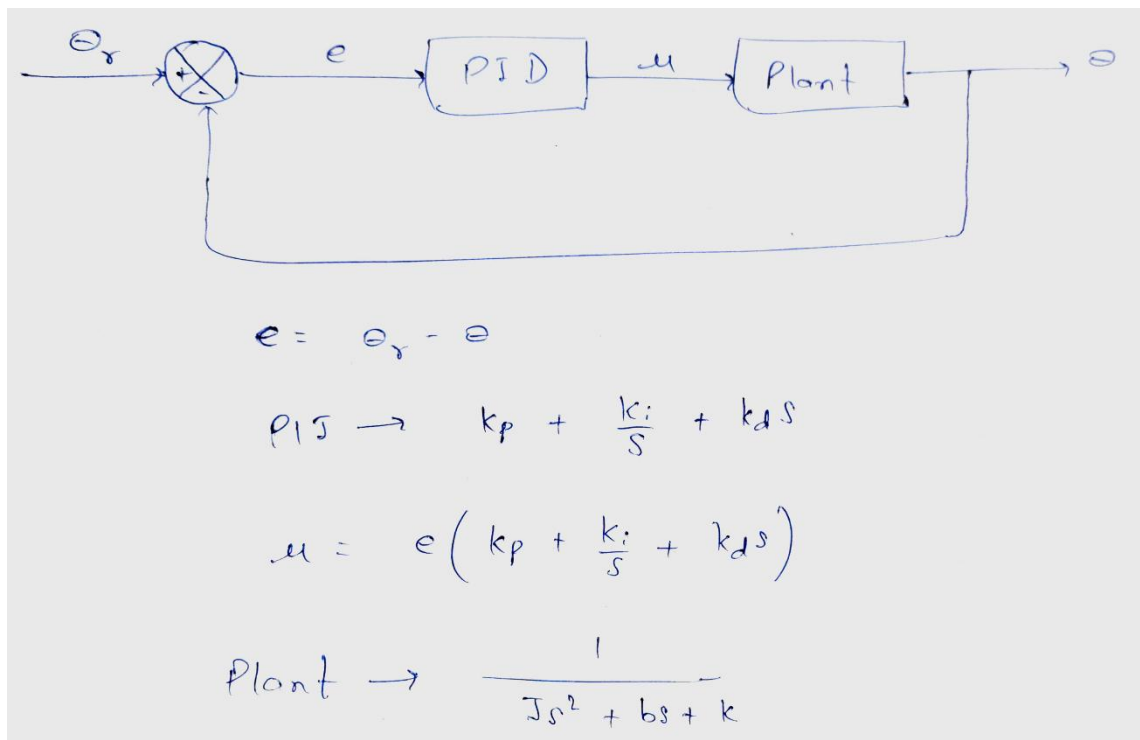


Fig. 4.4. Block Diagram of PID control

The plant contains the DC motor, gear system, lead screw and extruder.

When the PID loop starts Arduino continuously reads the current position using the hall-effect encoder attached to the DC motor shaft by counting the number of pulses received.

Since the motor is connected to a 3-stage planetary gear system, each motor revolution results in a very small, precise movement of the final output shaft.

PID programme set in Arduino-

```
// === Define pins ===
const int ENCODER_PIN_A = 2; // Interrupt pin
const int ENCODER_PIN_B = 4;
const int MOTOR_PWM = 10;    // ENA - PWM control
const int MOTOR_IN1 = 8;     // Direction pin
const int MOTOR_IN2 = 9;
```

It sets the connection pins in Arduino.

```
// === PID variables ===
float Kp = 1.5;
float Ki = 0.05;
float Kd = 0.0;
```

PID values set. We set the Kd derivative constant 0 because it adds noises in steady state condition.

```

    Serial.println("Enter desired angle in degrees:", Do);
    long desiredAngle = Do; // Target angle in degrees
    float currentAngle = 0;

```

Here we give the desired angle of rotation.

$$\text{Output Angle (degrees)} = \left(\frac{\text{Encoder Counts}}{\text{Counts per Motor Rev}} \right) \times \left(\frac{360^\circ}{\text{Gear Ratio}} \right)$$

```

void loop() {
    // === Calculate current angle ===
    currentAngle = (encoderCount / COUNTS_PER_REV) * 360.0;

    // === PID Calculations ===
    error = desiredAngle - currentAngle;
    integral += error;
    derivative = error - previousError;

    pidOutput = Kp * error + Ki * integral + Kd * derivative;

    previousError = error;

    // === Motor Control ===
    setMotor(pidOutput);

    Serial.print("Angle: "); Serial.print(currentAngle);
    Serial.print("   Error: "); Serial.print(error);
    Serial.print("   PWM: "); Serial.println(abs(pidOutput));

    delay(10);
}

```

The feed angle is converted into the no. of pulses from encoder required to reach at a desired position i.e. for one revolution 40500 pulses needed. Then Arduino reads the current number of pulses detected to find the current position.

Then a function, error counts the difference between the desired angle and the current angle.

Then the PID control function accumulate the error and calculate the output PWM signal for motor driver to set the desired speed of motor and to stop the motor.

$$\text{Control Output} = K_p \cdot \text{Error} + K_i \cdot \text{Integral} + K_d \cdot \text{Derivative}$$

```

void setMotor(float pidValue) {
    int pwm = constrain(abs(pidValue), 0, 255); // Limit PWM signal

    if (pidValue > 0) {
        digitalWrite(MOTOR_IN1, HIGH);
        digitalWrite(MOTOR_IN2, LOW);
    } else if (pidValue < 0) {
        digitalWrite(MOTOR_IN1, LOW);
        digitalWrite(MOTOR_IN2, HIGH);
    } else {
        digitalWrite(MOTOR_IN1, LOW);
        digitalWrite(MOTOR_IN2, LOW);
    }
}

```

This part of the code set the direction and speed of motor according to the PID value from the error.

Why PID Control is Important

In systems with high gear reduction, like a 3-stage planetary gear, the output moves very slowly and precisely. This is ideal for position control, but it also means:

- There can be delay between motor input and observable movement.
- System may overshoot or oscillate if not carefully controlled.
- High gear ratio causes increased inertia and mechanical resistance.

PID control continuously measures the **error** between the target position and the actual position (from encoder feedback) and adjusts the motor's speed and direction accordingly.

6. Further Aims

- **Experimental validation:** Conduct mechanical and biological testing to assess performance under simulated physiological loads.
- **Material innovations:** Explore alternative biocompatible materials to further reduce costs and improve durability.
- **Scalability and commercialization:** Develop a roadmap for large-scale production and clinical trials to ensure widespread adoption in India and other developing nations.

7. Conclusion

The development of a cost-effective internal bone-lengthening implant, as proposed in this project, presents a promising alternative to high-cost commercial systems like Fitbone and Precice Nail. By integrating a 6V DC motor, a three-stage planetary gear system, and a precision lead screw, we have demonstrated a mechanism capable of achieving smooth and controlled linear actuation suitable for distraction osteogenesis. The use of a PID-controlled closed-loop system—featuring Arduino Uno, L298N motor driver, and a Hall-effect encoder—enables highly accurate positioning, compensating for mechanical delays and non-linearities introduced by the gear reduction. This precise control is essential in medical applications, where overcorrection or instability can lead to complications in bone healing.

Our prototype effectively converts rotary motion into slow and measurable linear displacement, aligning well with the biological requirements of bone regeneration. The system achieves a controlled lengthening rate of approximately 0.3 mm per 2 minutes, which is consistent with clinical recommendations for gradual distraction. The modular design allows for wireless or mechanical external control, enhancing usability for both patients and physicians while reducing the need for invasive surgical adjustments.

Furthermore, the design focuses on affordability by leveraging domestic manufacturing, additive techniques, and open-source electronics, making the solution more accessible in resource-limited settings. While still in a developmental phase, the mechanical performance and theoretical outcomes indicate a strong potential for clinical applicability, pending experimental validation.

In summary, the project demonstrates a successful fusion of mechanical engineering, biomedical needs, and cost optimization. It lays the foundation for a future-ready orthopaedic solution that balances precision, functionality, and affordability—critical for transforming limb-lengthening treatment in developing healthcare systems.

8. References

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